Aggressiveness and brain amine concentration in dominant and subordinate finishing pigs fed the beta-adrenoreceptor agonist ractopamine.

Under farm conditions, aggression related to the formation of social hierarchy and competition for resources can be a major problem because of associated injuries, social stress, and carcass losses. Any factor that may affect the regulation and amount of aggression within a farmed system, for instance, feeding the beta-adrenoreceptor agonist ractopamine (RAC), is therefore worthy of investigation. The objectives of this study were to assess the effects of the widely used swine feed additive RAC, considering also the effects of sex and social rank on aggressiveness and concentrations of brain amines, neurotransmitters essential for controlling aggression, in finishing pigs. Thirty-two barrows and 32 gilts (4 pigs/pen by sex) were fed either a control diet or a diet with RAC (Paylean, Elanco Animal Health, Greenfield, IN) added (5 mg/kg for 2 wk, followed by 10 mg/kg for 2 wk). The top dominant and bottom subordinate pigs (16 pigs/sex) in each pen were determined after mixing by a 36-h period of continuous behavioral observation. These pigs were then subjected to resident-intruder tests (maximum 300 s) during the feeding trial to measure aggressiveness. At the end of wk 4, the amygdala, frontal cortex, hypothalamus, and raphe nuclei were dissected and analyzed for concentrations of dopamine (DA); serotonin (5-HT); their metabolites 3,4-dihydroxyphenyl acetic acid (DOPAC) and homovanillic acid, and 5-hydroxyindoleacetic acid (5-HIAA), respectively; norepinephrine; and epinephrine using HPLC. Ractopamine-fed gilts performed more attacks during the first 30 s of testing than pigs in all other subgroups ($P < 0.05$). By the end of the resident-intruder test (300 s), the dominant control gilts and barrows, and both dominant and subordinate RAC-fed gilts performed the greatest percentage of attacks ($P < 0.05$). Gilts had decreased norepinephrine and DOPAC concentrations in the amygdala and frontal cortex, and when fed RAC, gilts also had the least 5-HIAA concentration and greatest DA turnover rate in the amygdala ($P < 0.05$). The 5-HT concentration was less in the frontal cortex of gilts compared with barrows and in the raphe nuclei (single site for brain 5-HT synthesis) of dominant gilts ($P < 0.05$).

Ractopamine may be affecting aggressive behavior through indirect action on central regulatory mechanisms such as the DA system. The aggressive pattern observed in the tested pigs, especially in gilts, is likely linked to brain monoamine profiling of a deficient serotonergic system in the raphe nuclei, amygdala, and frontal cortex, and enhanced DA metabolism in the amygdala, brain areas vital for aggression regulation.